```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report

# Read dataset
data = pd.read_csv("spam.csv", encoding="latin1")

# Select message and category columns
data = data[["v1", "v2"]]

# Rename columns
data.columns = ["Category", "Message"]

# Convert categories into numbers
data["Category"] = data["Category"].map({
    "ham": 0,
    "spam": 1
})

# Separate input and output
X = data["Message"]
y = data["Category"]

# Convert text into numerical values
vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(X)

# Split dataset into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
# KNN Model
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)

y_pred_knn = knn.predict(X_test)

print("KNN Accuracy:", accuracy_score(y_test, y_pred_knn))
print("KNN Classification Report:")
print(classification_report(y_test, y_pred_knn))

# SVM Model
svm = SVC(kernel="linear")
svm.fit(X_train, y_train)

y_pred_svm = svm.predict(X_test)

print("SVM Accuracy:", accuracy_score(y_test, y_pred_svm))
print("SVM Classification Report:")
print(classification_report(y_test, y_pred_svm))
```